PAPER_695: NGC 7635 Bubble Nebula: O-Star Wind Driven Cavity and UQFF

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Abstract

NGC 7635 (Bubble Nebula) is a circumstellar nebula driven by the O-star BD+60°2522. The Weaver et al. (1977) analytic model for wind-blown cavities is applied with UQFF modifications to the terminal wind velocity and bubble expansion.

1. Parameters

Parameter	Value
M_*	$44M_\odot$
L_*	$4 \times 10^5 L_\odot$
$\dot{M}_{wind}$	$10^{-5} M_\odot$/yr
v_∞	2000 km/s
R_{bubble}	5 ly

2. Wind Luminosity

$$L_w = \frac{1}{2}\dot{M}v_\infty^2$$

3. Bubble Expansion (Weaver 1977)

$$R_b(t) = 0.88 \left(\frac{L_w}{\rho_0}\right)^{1/5} t^{3/5}$$

4. UQFF Wind Velocity

$$v_{UQFF} = v_\infty(1 + f_{TRZ})\sqrt{\frac{\rho_{UA}}{\rho_{SCm}}}$$

Enhanced by factor $\sqrt{\rho_{UA}/\rho_{SCm}} = \sqrt{10} \approx 3.16$.

5. Shock Compression (Strong Shock)

$$\frac{\rho_2}{\rho_1} = \frac{\gamma + 1}{\gamma - 1} = 4 \quad (\gamma = 5/3)$$

References

- Weaver et al. (1977), ApJ, 218, 377
- UQFF Framework, Star-Magic Session 174

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